STAT 185 - Homework #1

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Instructions

This assignment is intended to review your comfort with important ideas linear algebra and multivariable calculus and to assess your comfort with R coding and RMarkdown. Complete all problems below and compile your solutions into a single pdf document which must be submitted through Gradescope by Friday September 12th at 11:59 PM.

Notation

Throughout this assignment, all matrices are indicated in bold (e.g. $\mathbf{A}, \mathbf{B}$). All vectors are indicated with an arrow (e.g. $\vec{x}$) and should be interpreted as column vectors. Row vectors are viewed as transposed vectors (e.g. $\vec{x}^T$ is a row vector).

Linear Algebra and Calculus Review

Problem 1

Let $\vec{v}_1, \dots \vec{v}_k$ be vectors in $\mathbb{R}^d$.

a) Define the *span* of $\vec{v}_1, \dots \vec{v}_k$.

Answer: The span of the vectors $\vec{v}_1, \dots, \vec{v}_k$ is defined as the space of vectors that can be expressed as linear combinations of $\vec{v}_1, \dots, \vec{v}_k$. Mathematically,

$$\text{Span}(\vec{v}_1, \dots, \vec{v}_k) = \{c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_k\vec{v}_k \mid c_1, \dots, c_k \in \mathbb{R}\}$$

b) What does it mean for the vectors $\vec{v}_1, \dots, \vec{v}_k$ to be linearly independent?

Answer: Linear independence for the set of vectors $\vec{v}_1, \dots, \vec{v}_k$ means that each of the vectors cannot be expressed as a linear combination of the others. Mathematically, this means for any $\vec{v}_i$ in the set $\{\vec{v}_1, \dots, \vec{v}_k\}$,

$$\vec{v}_i \notin \text{Span}(\vec{v}_1, \dots, \vec{v}_{i-1}, \vec{v}_{i+1}, \dots, \vec{v}_k)$$

c) Assume that $k < d$. Let $\mathbf{V}$ be the $d \times k$ matrix with columns $\vec{v}_1, \dots, \vec{v}_k$. If $\vec{v}_1, \dots, \vec{v}_k$ are linearly independent, what can you say about the rank of $\mathbf{V}$.

Answer: The vectors of $\mathbf{V}$ span a k-dimensional vector space in $\mathbb{R}^d$. The rank of a matrix is the dimension of the column space. Since the k columns are linearly independent, the dimension of the column space is k.

$$\text{Rank}(\mathbf{V}) = k$$

Problem 2

Let $\vec{x} = (4, 5, 6)^T$ and the $\vec{w}_1 = (1, 1, 0)^T$, $\vec{w}_2 = (0, 0, 1)^T$, $\vec{w}_3 = (1, 0, 1)^T$ be vectors in $\mathbb{R}^3$.

a) Show that $\vec{w}_1$, $\vec{w}_2$, $\vec{w}_3$ are a basis for $\mathbb{R}^3$.

Answer: To show the vectors form a basis for $\mathbb{R}^3$, we must show they are linearly independent and span $\mathbb{R}^3$. Since we have 3 vectors in $\mathbb{R}^3$, we only need to show linear independence. We can do this by forming a

matrix $\mathbf{W}$ with these vectors as columns and showing it has full rank (i.e., it is invertible, and its reduced row echelon form is the identity matrix).

$$\mathbf{W} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

Attempting to reduce to row echelon form:

$$\begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - R_1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - R_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Since we are able to do Gaussian elimination upon $\mathbf{W}$ to get the identity matrix, the vectors of $\mathbf{W}$ are linearly independent and span a 3-dimensional space in $\mathbb{R}^3$. This is equivalent to saying the vectors are a basis for $\mathbb{R}^3$.

b) Find the coordinates of $\vec{x}$ is the basis defined by $\vec{w}_1$, $\vec{w}_2$, $\vec{w}_3$.

Answer: We want to find coefficients c_1, c_2, c_3 such that $c_1\vec{w}_1 + c_2\vec{w}_2 + c_3\vec{w}_3 = \vec{x}$. This can be solved by setting up an augmented matrix and finding the reduced row echelon form (RREF).

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 4 \\ 1 & 0 & 0 & 5 \\ 0 & 1 & 1 & 6 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 1 & 0 & 1 & 4 \\ 0 & 1 & 1 & 6 \end{array}\right] \xrightarrow{R_2 \leftarrow R_2 - R_1} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 0 & 1 & -1 \\ 0 & 1 & 1 & 6 \end{array}\right]$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & 1 & 6 \\ 0 & 0 & 1 & -1 \end{array}\right] \xrightarrow{R_2 \leftarrow R_2 - R_3} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -1 \end{array}\right]$$

Thus the coordinates of $\vec{x}$ in the basis defined by the vectors of W are $(5, 7, -1)^T$.

Problem 3

For the following subspace W of $\mathbb{R}^3$.

$$W = \{(z_1, z_2, z_3)^T \in \mathbb{R}^3 \mid z_1 - z_2 + z_3 = 0\}.$$

a) Find a matrix $\mathbf{P}$ such that $\mathbf{P}\vec{x}$ is the orthogonal projection of $\vec{x}$ onto the subspace W.

Answer: For the plane $z_1 - z_2 + z_3 = 0$, the normal vector is $\vec{n} = (1, -1, 1)^T$. The orthogonal projection matrix onto the subspace W (the plane) is given by the formula:

$$\mathbf{P} = \mathbf{I} - \frac{\vec{n}\vec{n}^T}{\vec{n}^T\vec{n}}$$

First, calculate $\vec{n}\vec{n}^T$:

$$\vec{n}\vec{n}^T = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$$

Next, calculate $\vec{n}^T\vec{n}$:

$$\vec{n}^T\vec{n} = (1)^2 + (-1)^2 + (1)^2 = 3$$

Now, construct $\mathbf{P}$:

$$\mathbf{P} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{1}{3}\begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 2/3 & 1/3 & -1/3 \\ 1/3 & 2/3 & 1/3 \\ -1/3 & 1/3 & 2/3 \end{bmatrix}$$

b) Using the matrix from (a), find the projection of $(2, 1, 2)^T$ onto W.

Answer: Let $\vec{x} = (2, 1, 2)^T$. We find the projection by calculating $\mathbf{P}\vec{x}$.

$$\mathbf{P}\vec{x} = \begin{bmatrix} 2/3 & 1/3 & -1/3 \\ 1/3 & 2/3 & 1/3 \\ -1/3 & 1/3 & 2/3 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} (4/3) + (1/3) - (2/3) \\ (2/3) + (2/3) + (2/3) \\ (-2/3) + (1/3) + (4/3) \end{bmatrix} = \begin{bmatrix} 3/3 \\ 6/3 \\ 3/3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

Thus $(1, 2, 1)^T$ is the projection onto the space W.

Problem 4

Consider the helix in $\mathbb{R}^3$ parameterized by the equation

$$t \mapsto \vec{r}(t) = \begin{bmatrix} \cos t \\ \sin t \\ t \end{bmatrix} \qquad t \in (-\infty, \infty).$$

Find the unit tangent vector and the unit normal vector at the point corresponding to $t = \pi/2$.

Answer: To find the unit tangent vector, we need to get the derivative of $\vec{r}(t)$ with respect to t.

$$\frac{d\vec{r}}{dt} = \begin{bmatrix} -\sin t \\ \cos t \\ 1 \end{bmatrix}$$

Thus the slope (tangent vector) at $t = \pi/2$ is:

$$\vec{r}'(\pi/2) = \begin{bmatrix} -\sin(\pi/2) \\ \cos(\pi/2) \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

Normalizing this vector to find the unit tangent vector $\vec{T}$:

$$||\vec{r}'(\pi/2)|| = \sqrt{(-1)^2 + 0^2 + 1^2} = \sqrt{2}$$

$$\vec{T}(\pi/2) = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} -\sqrt{2}/2 \\ 0 \\ \sqrt{2}/2 \end{bmatrix}$$

The unit normal vector is orthogonal to the unit tangent vector. Thus their dot product is 0.

$$\begin{bmatrix} -\sqrt{2}/2 \\ 0 \\ \sqrt{2}/2 \end{bmatrix} \cdot \vec{x} = 0$$

One such vector is $\vec{x} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$. Normalizing this vector gives a unit normal vector $\vec{N}$:

$$||\vec{x}|| = \sqrt{1^2 + 0^2 + 1^2} = \sqrt{2}$$

$$\vec{N}(\pi/2) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} \sqrt{2}/2 \\ 0 \\ \sqrt{2}/2 \end{bmatrix}$$

Problem 5

Consider the helix from problem 4. Find the arclength of the path along the helix connecting points $(1,0,0)^T$ and $(1,0,2\pi)^T$.

Answer: The arc length L is given by the integral $L = \int_a^b ||\frac{d\vec{r}}{dt}||dt$. The position on our curve is $\vec{r}(t) = (\cos t, \sin t, t)^T$. The positions t of the endpoints are: For $(1,0,0)^T \implies t = 0$. For $(1,0,2\pi)^T \implies t = 2\pi$. As previously stated, the derivative of $\vec{r}(t)$ is $\frac{d\vec{r}}{dt} = \begin{bmatrix} -\sin t \\ \cos t \\ 1 \end{bmatrix}$. The magnitude of this is $||\frac{d\vec{r}}{dt}|| = \sqrt{(-\sin t)^2 + (\cos t)^2 + 1^2} = \sqrt{\sin^2 t + \cos^2 t + 1} = \sqrt{1+1} = \sqrt{2}$.

$$L = \int_0^{2\pi} \sqrt{2}dt = [\sqrt{2}t]_0^{2\pi} = \sqrt{2}(2\pi) - 0 = 2\pi\sqrt{2}$$

The arc length is $2\pi\sqrt{2}$.

Problem 6

Let $\mathbf{A}$ be a $m \times n$ real-valued matrix. Show that $\mathbf{A}\mathbf{A}^T$ is positive semi-definite (all its eigenvalues are real and nonnegative).

Answer: A matrix $\mathbf{M}$ is positive semi-definite if $\vec{x}^T\mathbf{M}\vec{x} \geq 0$ for all non-zero vectors $\vec{x}$. We must show that $\vec{x}^T(\mathbf{A}\mathbf{A}^T)\vec{x} \geq 0$ for all $\vec{x} \in \mathbb{R}^m$. Rewrite the expression:

$$\vec{x}^T\mathbf{A}\mathbf{A}^T\vec{x} = (\vec{x}^T\mathbf{A})(\mathbf{A}^T\vec{x}) = (\mathbf{A}^T\vec{x})^T(\mathbf{A}^T\vec{x})$$

Let $\vec{y} = \mathbf{A}^T\vec{x}$. The expression becomes $\vec{y}^T\vec{y}$. This is the dot product of a vector with itself, which is equivalent to the squared Euclidean norm of the vector, $||\vec{y}||^2$.

$$\vec{y}^T\vec{y} = \sum_{i=1}^{n} y_i^2 = ||\vec{y}||^2$$

The sum of squares is always greater than or equal to zero. Thus, $\vec{x}^T\mathbf{A}\mathbf{A}^T\vec{x} \geq 0$, so the expression holds and $\mathbf{A}\mathbf{A}^T$ is Positive Semi-Definite (PSD).

Problem 7

Let A be an $m \times m$ matrix with constant entries $A_{ij} = a_{ij}$ for $1 \leq i \leq j \leq m$ and $\vec{x} = (x_1, \ldots, x_m)^T$ be a vector in $\mathbb{R}^m$.

a) Calculate

$$\nabla\left(\vec{x}^T A\vec{x}\right)$$

where $\nabla = \left(\frac{\partial}{\partial x_1}, \ldots, \frac{\partial}{\partial x_m}\right)^T$. *For full credit, your final answer cannot contain any summation notation.*

Answer: This is the quadratic form. It is known that the derivative of the quadratic form is $\nabla(\vec{x}^T\mathbf{A}\vec{x}) = (\mathbf{A} + \mathbf{A}^T)\vec{x}$. For completeness, I will show this is true. Let $f(\vec{x}) = \vec{x}^T\mathbf{A}\vec{x} = \sum_{i=1}^m \sum_{j=1}^m x_i A_{ij} x_j$. We need to find the partial derivative with respect to each x_k.

$$\frac{\partial f(\vec{x})}{\partial x_k} = \frac{\partial}{\partial x_k}\left(\sum_{i=1}^{m}\sum_{j=1}^{m} x_i A_{ij} x_j\right)$$

Using the product rule, the derivative is non-zero only for terms where $i = k$ or $j = k$.

$$\frac{\partial f(\vec{x})}{\partial x_k} = \sum_{j=1}^{m} A_{kj}x_j + \sum_{i=1}^{m} x_i A_{ik}$$

The first term, $\sum_{j=1}^{m} A_{kj}x_j$, is the k-th element of the vector $\mathbf{A}\vec{x}$. The second term, $\sum_{i=1}^{m} x_i A_{ik} = \sum_{i=1}^{m} A_{ik}^T x_i$, is the k-th element of the vector $\mathbf{A}^T\vec{x}$. So the k-th component of the gradient is $(\mathbf{A}\vec{x})_k + (\mathbf{A}^T\vec{x})_k = ((\mathbf{A} + \mathbf{A}^T)\vec{x})_k$. Thus, the gradient vector is:

$$\nabla(\vec{x}^T\mathbf{A}\vec{x}) = (\mathbf{A} + \mathbf{A}^T)\vec{x}$$

b) Calculate the Hessian matrix H where

$$H_{ij} = \frac{\partial^2}{\partial x_i \partial x_j}\left(\vec{x}^T A\vec{x}\right).$$

For full credit your answer cannot contain any summation notation.

Answer: We take the second derivative.

$$H_{ij} = \frac{\partial}{\partial x_i}\left(\frac{\partial}{\partial x_j}(\vec{x}^T\mathbf{A}\vec{x})\right) = \frac{\partial}{\partial x_i}\left[(\mathbf{A} + \mathbf{A}^T)\vec{x}\right]_j$$

The j-th component of $(\mathbf{A} + \mathbf{A}^T)\vec{x}$ is $\sum_{k=1}^{m}(A_{jk} + A_{kj})x_k$.

$$H_{ij} = \frac{\partial}{\partial x_i}\left(\sum_{k=1}^{m}(A_{jk} + A_{kj})x_k\right)$$

The derivative with respect to x_i is non-zero only when $k = i$.

$$H_{ij} = A_{ji} + A_{ij}$$

This is the (i, j)-th entry of the matrix $\mathbf{A}^T + \mathbf{A}$. Thus,

$$\mathbf{H} = \mathbf{A} + \mathbf{A}^T$$

c) Let ∇_A denote the matrix of partial derivatives with respect to the entries of $\mathbf{A}$, i.e. $(\nabla_A)_{ij} = \partial/\partial a_{ij}$. Find $\nabla_A(\vec{x}^T\mathbf{A}\vec{x})$ (this is the matrix where each entry corresponds to the partial derivative of $\vec{x}^T\mathbf{A}\vec{x}$ with respect to the corresponding entry of $\mathbf{A}$. *For full credit your answer cannot contain any summation notation.*

Answer: We are finding $\nabla_A(\vec{x}^T\mathbf{A}\vec{x})$. First, write out the quadratic form in summation notation.

$$\vec{x}^T\mathbf{A}\vec{x} = \sum_{i=1}^{m}\sum_{j=1}^{m} x_i a_{ij} x_j$$

Now we take the partial derivative with respect to an arbitrary entry a_{kl} of the matrix $\mathbf{A}$.

$$\frac{\partial}{\partial a_{kl}}(\vec{x}^T\mathbf{A}\vec{x}) = \frac{\partial}{\partial a_{kl}}\left(\sum_{i=1}^{m}\sum_{j=1}^{m} x_i a_{ij} x_j\right)$$

The derivative is zero for all terms in the summation except for the one where $i = k$ and $j = l$.

$$\frac{\partial}{\partial a_{kl}}(x_k a_{kl} x_l) = x_k x_l$$

This is the entry in the k-th row and l-th column of the resulting matrix. The matrix whose (k, l)-th entry is $x_k x_l$ is the outer product of $\vec{x}$ with itself.

$$\nabla_A(\vec{x}^T\mathbf{A}\vec{x}) = \vec{x}\vec{x}^T$$

R Coding

These problems are intended to assess your comfort with R syntax, loading packages and using associated help files, interacting with data frames, and generating figures.

Problem 1

Later in the class, we'll use the package *dimRed* to implement several manifold learning methods. You will need to install *dimRed* from CRAN and load the package for this problem. Use the documentation for *dimRed* to find commands to list i) all available dimension reduction methods and ii) all data sets which come with the package.

```
# Install and load the dimRed package if not already installed
if (!require("dimRed")) {
  install.packages("dimRed")
}
library(dimRed)

# List all available dimensionality reduction methods
print("Available dimension reduction methods:")
```

```
## [1] "Available dimension reduction methods:"
```

```
print(dimRedMethodList())
```

```
##  [1] "DiffusionMaps"       "DRR"                 "FastICA"
##  [4] "KamadaKawai"         "DrL"                 "FruchtermanReingold"
##  [7] "HLLE"                "Isomap"              "kPCA"
## [10] "PCA_L1"              "MDS"                 "nMDS"
## [13] "NNMF"                "PCA"                 "tSNE"
## [16] "UMAP"
```

Problem 2

The following script will download the MovieLens 10M dataset containing 10 million ratings of 10 thousand movies from 70 thousand reviewers. The data frame *movielens* contains all ratings in the dataset.

```
dl <- tempfile()
download.file("http://files.grouplens.org/datasets/movielens/ml-10m.zip", dl)
ratings <- fread(text = gsub("::", "\t",
                             readLines(unzip(dl, "ml-10M100K/ratings.dat"))),
                 col.names = c("userId", "movieId", "rating", "timestamp"))
movies <- str_split_fixed(readLines(unzip(dl, "ml-10M100K/movies.dat")), "\\::", 3)
colnames(movies) <- c("movieId", "title", "genres")
movies <- as.data.frame(movies) %>%
  mutate(movieId = as.numeric(movieId),
         title = as.character(title),
         genres = as.character(genres))
movielens <- left_join(ratings, movies, by = "movieId")
rm(movies,ratings,dl)
```

a) The statements regarding the number of movies and reviewers are only approximate. What are the exact number of movies and reviewers in the dataset?

```
num_reviewers <- n_distinct(movielens$userId)

num_movies <- n_distinct(movielens$movieId)

cat("Exact number of reviewers:", num_reviewers, "\n")
```

```
## Exact number of reviewers: 69878
```

```r
cat("Exact number of movies:", num_movies, "\n")
```

```
## Exact number of movies: 10677
```

b) Remove all movies which have received 100 or fewer reviews. How many movies are left in the dataset?

```r
movie_review_counts <- movielens %>%
  group_by(movieId) %>%
  summarise(num_reviews = n())

movies_to_keep <- movie_review_counts %>%
  filter(num_reviews > 100)

movielens_filtered <- movielens %>%
  semi_join(movies_to_keep, by = "movieId")

num_movies_left <- n_distinct(movielens_filtered$movieId)

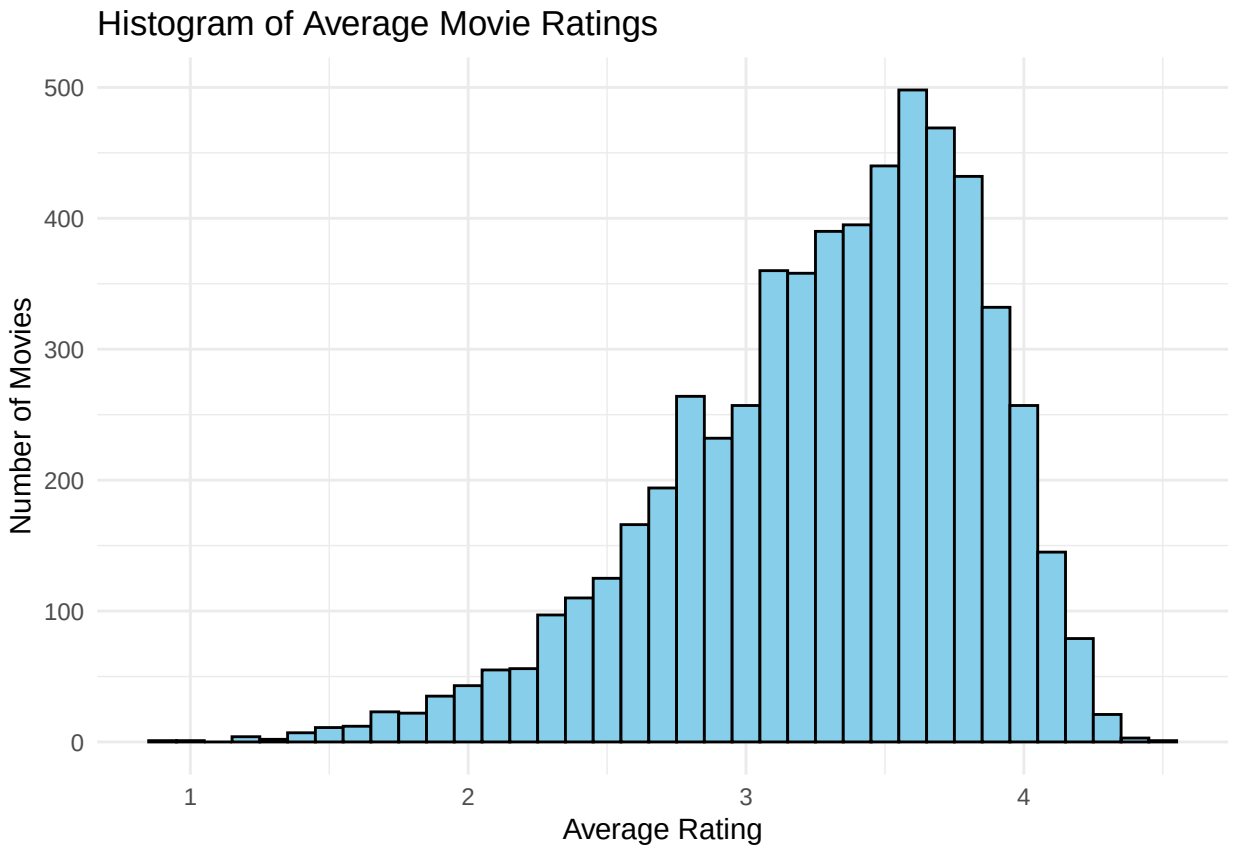
cat("Number of movies left after", num_movies_left, "\n")
```

```
## Number of movies left after 5897
```

c) Using the filtered dataset from part (b), calculate the average rating of each movie. Plot a histogram of the average ratings.

```r
# claculate
avg_movie_ratings <- movielens_filtered %>%
  group_by(movieId, title) %>%
  summarise(avg_rating = mean(rating, na.rm = TRUE), .groups = 'drop')

# plot
ggplot(avg_movie_ratings, aes(x = avg_rating)) +
  geom_histogram(binwidth = 0.1, color = "black", fill = "skyblue") +
  labs(title = "Histogram of Average Movie Ratings",
       x = "Average Rating",
       y = "Number of Movies") +
  theme_minimal()
```



d) Based on the results above, which movie(s) had the highest and lowest average rating and what were the ratings?

```
highest_rating <- max(avg_movie_ratings$avg_rating, na.rm = TRUE)
highest_rated_movies <- avg_movie_ratings %>%
  filter(avg_rating == highest_rating)

lowest_rating <- min(avg_movie_ratings$avg_rating, na.rm = TRUE)
lowest_rated_movies <- avg_movie_ratings %>%
  filter(avg_rating == lowest_rating)

cat("Movie(s) with the highest average rating:\n")
```

```
## Movie(s) with the highest average rating:
```

```
print(highest_rated_movies)
```

```
## # A tibble: 1 x 3
##   movieId title                              avg_rating
##     <dbl> <chr>                                   <dbl>
## 1     318 Shawshank Redemption, The (1994)         4.46
```

```
cat("\nMovie(s) with the lowest average rating:\n")
```

```
##
## Movie(s) with the lowest average rating:
```

```
print(lowest_rated_movies)
```

```
## # A tibble: 1 x 3
```

```
##   movieId title                         avg_rating
##     <dbl> <chr>                              <dbl>
## 1    6483 From Justin to Kelly (2003)        0.924
```